
Applied Cyber Threat Analysis – Cryptography Lab

- This lab was conducted within the Virginia Cyber Range Environment.
 - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
 - Cheche Agada. (2024). Cryptography Lab IT-462: Applied Cyber Threat Analysis, George Mason University
-

Introduction:

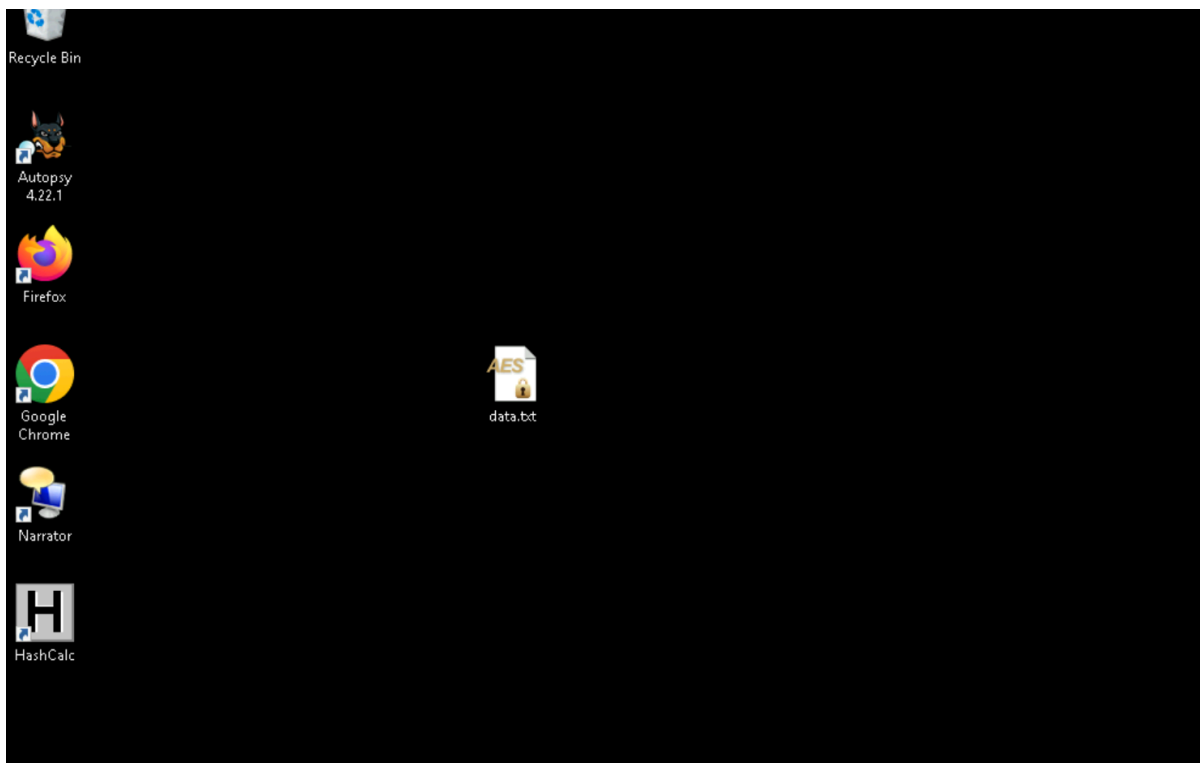
Cryptography is the art of secret writing. It is the process of using mathematical algorithms to protect information from unauthorized access. In this exercise, we will demonstrate the use of cryptography to protect information.

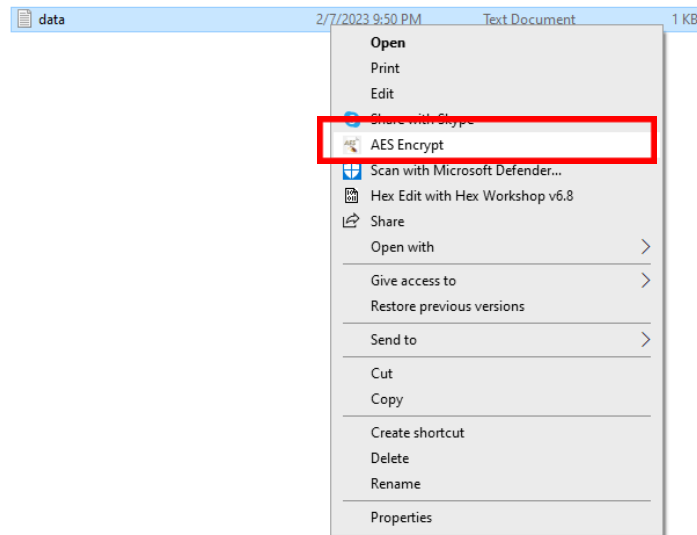
Objective:

Demonstrate the use of several cryptographic algorithms for information protection.

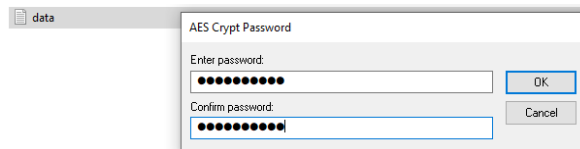
Task 1A – Symmetric key encryption using AES on Windows (GUI)

- Download and install AesCrypt (<https://www.aescrypt.com/download/>)
- Set the path (i.e. C:\Program Files\AesCrypt\) in your system variables so AesCrypt is accessible from anywhere on your system. (Both steps should have been done in week 1)
- Create a file named 'data.txt', make sure the file contains some data (you may use any file, it does not have to be a text file)
- To encrypt a file in Windows GUI, right click on the file you want to encrypt (i.e. data.txt) and select 'AES Encrypt'

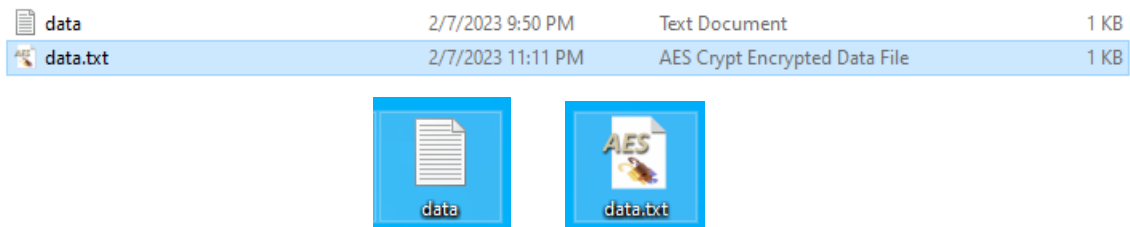




- You will be prompted to provide and confirm your encryption password.



- Once a password is confirmed, AES creates a new file (i.e data.txt.aes). This is the encrypted file.



- To decrypt the file, double click on it and provide the password used for encryption (**Make sure the original file is removed before decryption to prevent name collision**).

Task 1B – Symmetric key encryption using AES on Windows (command line)

- Open a command prompt window and navigate to the location of the file you want to encrypt.
- To encrypt, enter the following command on the console `aesencrypt -e -p password data.txt` (**-e** and **-p** indicate an encryption operation and encryption password respectively).

```

CA: Command Prompt

C:\Users\oagada\Desktop>aescript -e -p lasgidi678 data.txt
C:\Users\oagada\Desktop>aescript -d -p lasgidi678 data.txt.aes

```

- To decrypt, enter the following command on the console `aescript -d -p password data.txt.aes` (-d and -p indicate a decryption operation and decryption password respectively).
- Make sure the original file is removed before decryption to prevent name collision.

```

Administrator: Command Prompt

C:\Users\student>cd Desktop

C:\Users\student\Desktop>dir\
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

Directory of C:\

11/14/2018  06:56 AM  <DIR>          EFI
05/13/2020  05:58 PM  <DIR>          PerfLogs
06/08/2025  06:37 PM  <DIR>          Program Files
06/02/2025  01:18 AM  <DIR>          Program Files (x86)
06/02/2025  12:09 AM  <DIR>          Users
05/28/2025  05:18 PM  <DIR>          Windows
               0 File(s)              0 bytes
               6 Dir(s)  3,934,126,080 bytes free

C:\Users\student\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

Directory of C:\Users\student\Desktop

06/08/2025  08:22 PM  <DIR>          .
06/08/2025  08:22 PM  <DIR>          ..
06/08/2025  08:20 PM                6 data.txt
06/02/2025  01:18 AM          976 HashCalc.lnk
               2 File(s)              982 bytes
               2 Dir(s)  3,934,126,080 bytes free

```

```

Administrator: Command Prompt

C:\Users\student\Desktop>aescript -e -p tfracyon data.txt
Encrypting: data.txt

C:\Users\student\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

Directory of C:\Users\student\Desktop

06/08/2025  08:25 PM  <DIR>          .
06/08/2025  08:25 PM  <DIR>          ..
06/08/2025  08:20 PM                6 data.txt
06/08/2025  08:25 PM          188 data.txt.aes
06/02/2025  01:18 AM          976 HashCalc.lnk
               3 File(s)              1,170 bytes
               2 Dir(s)  3,934,126,080 bytes free

C:\Users\student\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

Directory of C:\Users\student\Desktop

06/08/2025  08:25 PM  <DIR>          .
06/08/2025  08:25 PM  <DIR>          ..
06/08/2025  08:25 PM          188 data.txt.aes
06/02/2025  01:18 AM          976 HashCalc.lnk
               2 File(s)              1,164 bytes
               2 Dir(s)  3,934,126,080 bytes free

```

```

Administrator: Command Prompt

Directory of C:\Users\student\Desktop

06/08/2025  08:25 PM  <DIR>          .
06/08/2025  08:25 PM  <DIR>          ..
06/08/2025  08:25 PM          188 data.txt.aes
06/02/2025  01:18 AM          976 HashCalc.lnk
               2 File(s)              1,164 bytes
               2 Dir(s)  3,934,126,080 bytes free

C:\Users\student\Desktop>aescript -d -p tfracyon data.txt
Input file does not end with .aes and no output file was specified: data.txt

C:\Users\student\Desktop>aescript -d -p tfracyon data.txt.aes
Decrypting: data.txt.aes

C:\Users\student\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

Directory of C:\Users\student\Desktop

06/08/2025  08:26 PM  <DIR>          .
06/08/2025  08:26 PM  <DIR>          ..
06/08/2025  08:26 PM                6 data.txt
06/08/2025  08:25 PM          188 data.txt.aes
06/02/2025  01:18 AM          976 HashCalc.lnk
               3 File(s)              1,170 bytes
               2 Dir(s)  3,934,126,080 bytes free

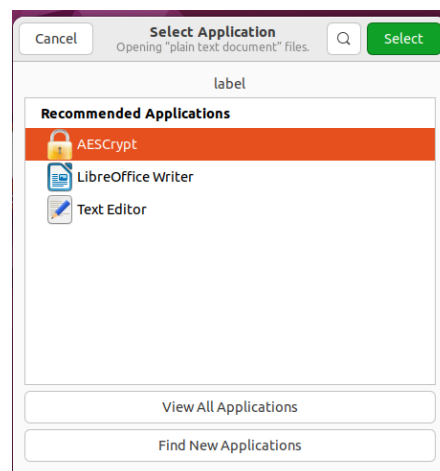
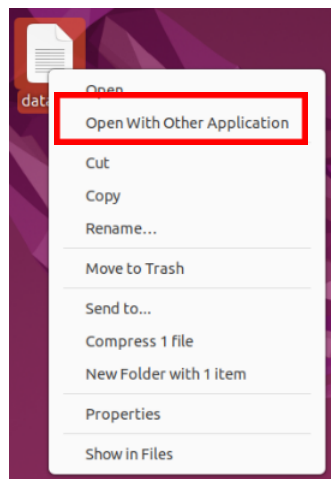
```

Task 2A – Symmetric key encryption using AES on Linux Ubuntu (GUI)

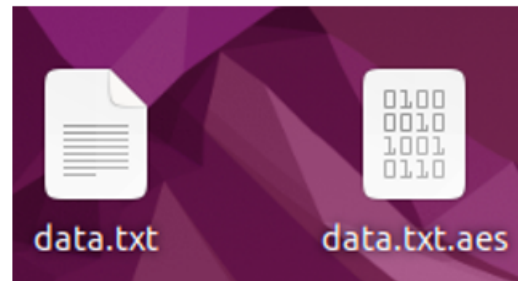
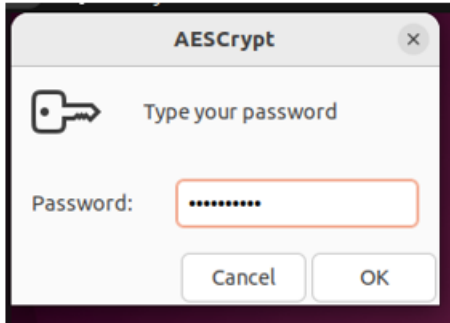
- Download [AES Crypt - GUI \(Linux 64-bit\)](#) (This should have been done in week 1).
- Select the debian (deb) Desktop & Command-Line version.
- Fire up a terminal window, navigate to the location of the downloaded file.
- Install the application using the following command:
`“sudo dpkg -i aescrypt_gui-4.0.4-Linux-x86_64.deb”`.

```
student@desktop:~$ aescrypt
aescrypt: command not found
student@desktop:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos thinclient_drives
student@desktop:~$ cd Downloads/
student@desktop:~/Downloads$ ls
aescrypt_gui-4.0.4-Linux-x86_64.deb
student@desktop:~/Downloads$ aescrypt
aescrypt: command not found
student@desktop:~/Downloads$ sudo dpkg -i aescrypt_gui-4.0.4-Linux-x86_64.deb
Selecting previously unselected package aescrypt_gui.
(Reading database ... 186933 files and directories currently installed.)
Preparing to unpack aescrypt_gui-4.0.4-Linux-x86_64.deb ...
Unpacking aescrypt_gui (4.0.4) ...
Setting up aescrypt_gui (4.0.4) ...
Processing triggers for mailcap (3.70+nmlubuntu1) ...
Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu1) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for shared-mime-info (2.1-2) ...
student@desktop:~/Downloads$ aescrypt
Specify either encrypt (-e), decrypt (-d), or generate (-g) mode
student@desktop:~/Downloads$
```

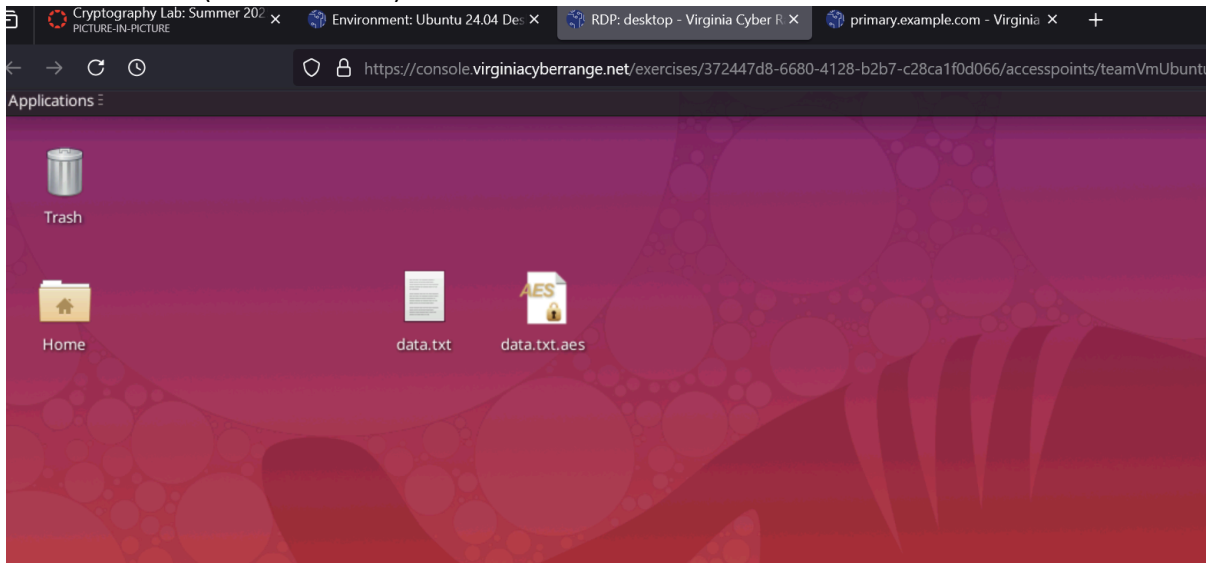
- To encrypt a file, right click on the target file and select ‘Open with other Application’. Then select AESCrypt from the dialog box.



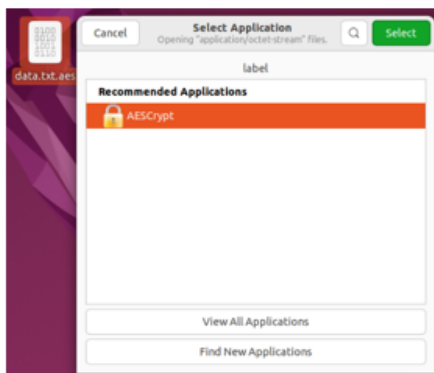
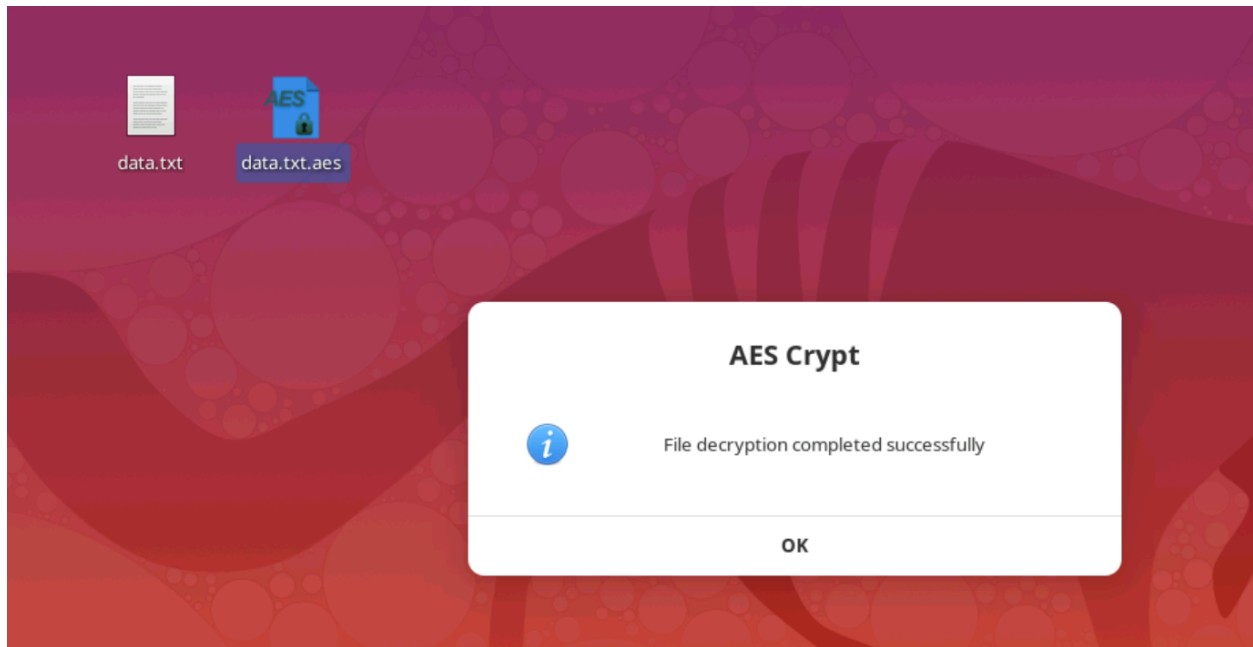
- Provide and confirm your encryption password.



- A new file (i.e. data.txt.aes) will be created.



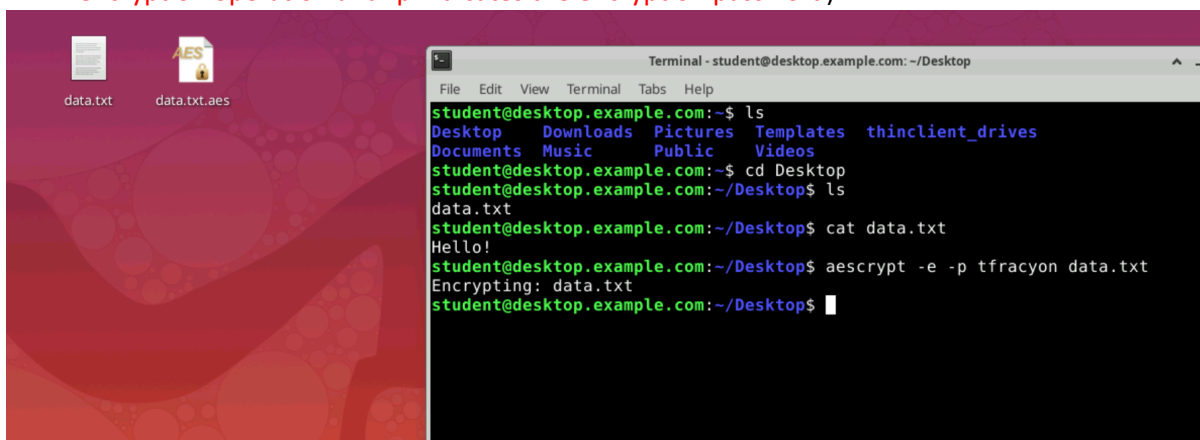
- To decrypt a file, right click on the target file and select AESCrypt (remove the original file to avoid error due to name collision).
- Provide (and confirm) the same password used for encryption.
- The encrypted file will be decrypted.



Task 2 B – Symmetric key encryption using AES on Linux Ubuntu command line

NOTE: Ensure AESCrypt is properly installed as described in Task 2A.

- Identify the file to be encrypted and first display the content of the file on the command line using the cat command (l. e. `cat data.txt`).
- To encrypt a file, use the following command: `aescrypt -e -p lasgidi678 data.txt` (-e indicates an encryption operation and -p indicates the encryption password).



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- A new file is created (i.e. data.txt.aes).

```
root@cheche-virtual-machine:/home/cheche/Desktop# ls
data.txt  pyAesCrypt
root@cheche-virtual-machine:/home/cheche/Desktop# cat data.txt
This is my test file
This is my test file
This is my test file
This is my test file
This is my test file
root@cheche-virtual-machine:/home/cheche/Desktop# aescrypt -e -p lasgidi678 data.txt
root@cheche-virtual-machine:/home/cheche/Desktop# ls
data.txt  data.txt.aes  pyAesCrypt
root@cheche-virtual-machine:/home/cheche/Desktop# cat data.txt.aes
AESCREATED_BYaescrypt 3.11***
*****2jQ*(H"V-6填F9**
}***
**:@m4YJgFHwLluoNoo1oTNleoX
Ofw**
le+**($**Y*\j***I?*****,/p**eUL?o***5o`***8Z*G**"***H6=,***M***XQ***n***4e`H+qae!  *jeb*****aeKoj*****U**\9root@cheche-virtual-machine:/home/cheche/Desktop#
```

Class Exercise 1

- Display the content of the new file using the cat command (i.e. `cat data.txt.aes`). What do you see? Explain.

```
Encrypting: data.txt
student@desktop.example.com:~/Desktop$ cat data.txt.aes
{c00y0r0"0}0L0>'oc0?00T10E7D":z0W0-~4+60Fu0student@desktop.example.com:~/Desktop
$
```

The data inside the file has been encrypted. What we see is a hashed string.

- Using the ubuntu command line, decrypt the file using the decryption command (i.e. `aescrypt -d -p lasgidi678 data.txt.aes`)
- Remember to remove the original file to avoid error due to name collision.

```
File Edit View Terminal Tabs Help
student@desktop.example.com:~/Desktop$ cat data.txt.aes
Encrypting: data.txt
student@desktop.example.com:~/Desktop$ ls
data.txt  data.txt.aes
student@desktop.example.com:~/Desktop$
```

Task 3A – Symmetric key encryption using AES and Python on Windows

- Install and configure the latest version of python 3. Set the path in the system/environment variables so you can run python from anywhere on the system.
- On the command prompt, install pyAesCrypt using the python package installer, pip (i.e. `pip install pyAesCrypt`) (Both steps should have been done in week 1)

Command Prompt

```
C:\Users\oagada\Desktop>pip install pyaescrypt
Collecting pyaescrypt
  Using cached pyAesCrypt-6.0.0-py3-none-any.whl (15 kB)
Requirement already satisfied: cryptography in c:\users\oagada\appdata\local\programs\python\python311\lib\site-packages (from pyaescrypt) (39.0.1)
Requirement already satisfied: cffi>=1.12 in c:\users\oagada\appdata\local\programs\python\python311\lib\site-packages (from cryptography->pyaescrypt) (1.15.1)
Requirement already satisfied: pycparser in c:\users\oagada\appdata\local\programs\python\python311\lib\site-packages (from cffi>=1.12->cryptography->pyaescrypt) (2.21)
Installing collected packages: pyaescrypt
Successfully installed pyaescrypt-6.0.0
```

- Open a text editor like **notepad++**, **notepad**, or **Python IDLE** (preferred) and type the code as shown in the screenshot below.
- Save the file as a python program, you may call it any name, but the file extension must be `.py` (i.e. `pyAes_Crypt.py`)
- Ensure the program is in the same directory as the file you want to encrypt. Also, note the parameters provided in the encryption function.
- To encrypt a file, run the Python program (i.e. `python pyAes_Crypt.py`)

```
*pyAes_Crypt.py - C:\Users\oagada\Desktop\pyAes_Crypt.py (3.11.1)*
File Edit Format Run Options Window Help

import pyAesCrypt as crypto

password = 'lasgidi678'

##Encrypt
crypto.encryptFile('data.txt', 'data.txt.aes', password)

##Decrypt
crypto.decryptFile('data.txt.aes', 'new_data.txt', password)
```

NOTE:

- In the screenshot, `'data.txt'` is the name of the file to be encrypted, `'data.txt.aes'` would be the name of the encrypted file, and `'password'`, is the encryption password.
- This is a simple program that only carries out one operation at a time, encryption or decryption, but not both. To decrypt, comment out the encryption function and uncomment the decryption function.
- Remember to remove the original file to avoid error due to name collision or use a different file name.

```
Administrator: Command Prompt

C:\Users\student\Desktop>dir
Volume in drive C has no label.
Volume Serial Number is 5EC2-6FD4

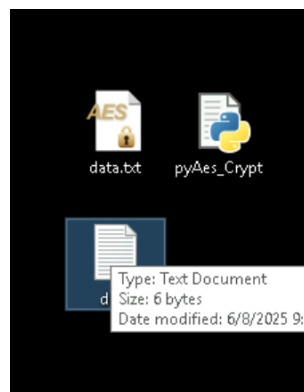
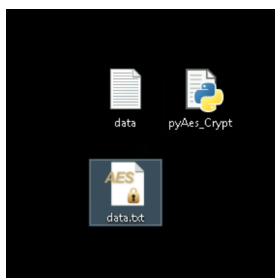
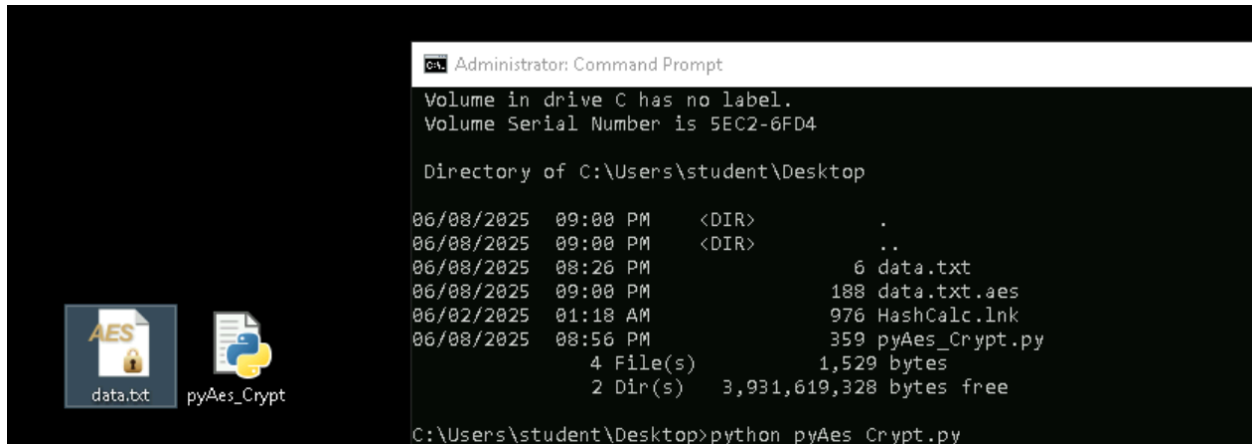
Directory of C:\Users\student\Desktop

06/08/2025  08:59 PM    <DIR>          .
06/08/2025  08:59 PM    <DIR>          ..
06/08/2025  08:26 PM                6 data.txt
06/02/2025  01:18 AM            976 HashCalc.lnk
06/08/2025  08:56 PM           359 pyAes_Crypt.py
               3 File(s)            1,341 bytes
               2 Dir(s)  3,931,308,032 bytes free

C:\Users\student\Desktop>python pyAes_Crypt.py
```


Class Exercise 2

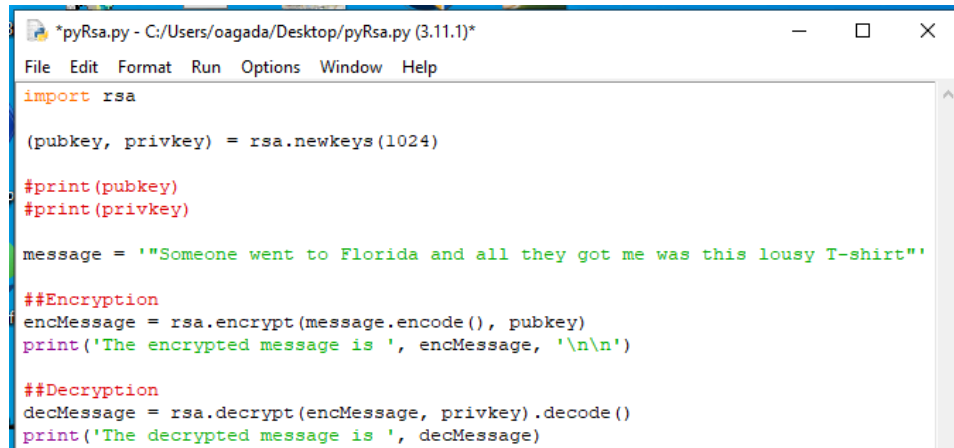
- Use the above program to decrypt the encrypted file. Provide screen shots to show your work.

**Task 3B – Symmetric key encryption using AES and Python on Ubuntu****Class Exercise 3**

- Use the python program in task 3A to encrypt a file in Ubuntu.
- Ubuntu usually comes with python 3 preinstalled, however you may have to install the python package installer, pip (i.e. `sudo apt install python3-pip`). (This should have been done in wk 1)
- Show your steps and include screenshots.

Task 4A – Asymmetric key encryption using RSA and Python on Ubuntu

- Open a text editor like **notepad++**, **notepad**, or **Python IDLE** (preferred) and type the code as shown in the screenshot below.
- Save the file as a python program, you may call it any name, but the file extension must be `‘.py’` (i.e. `pyRsa.py`)



```
*pyRsa.py - C:/Users/oagada/Desktop/pyRsa.py (3.11.1)*
File Edit Format Run Options Window Help
import rsa

(pubkey, privkey) = rsa.newkeys(1024)

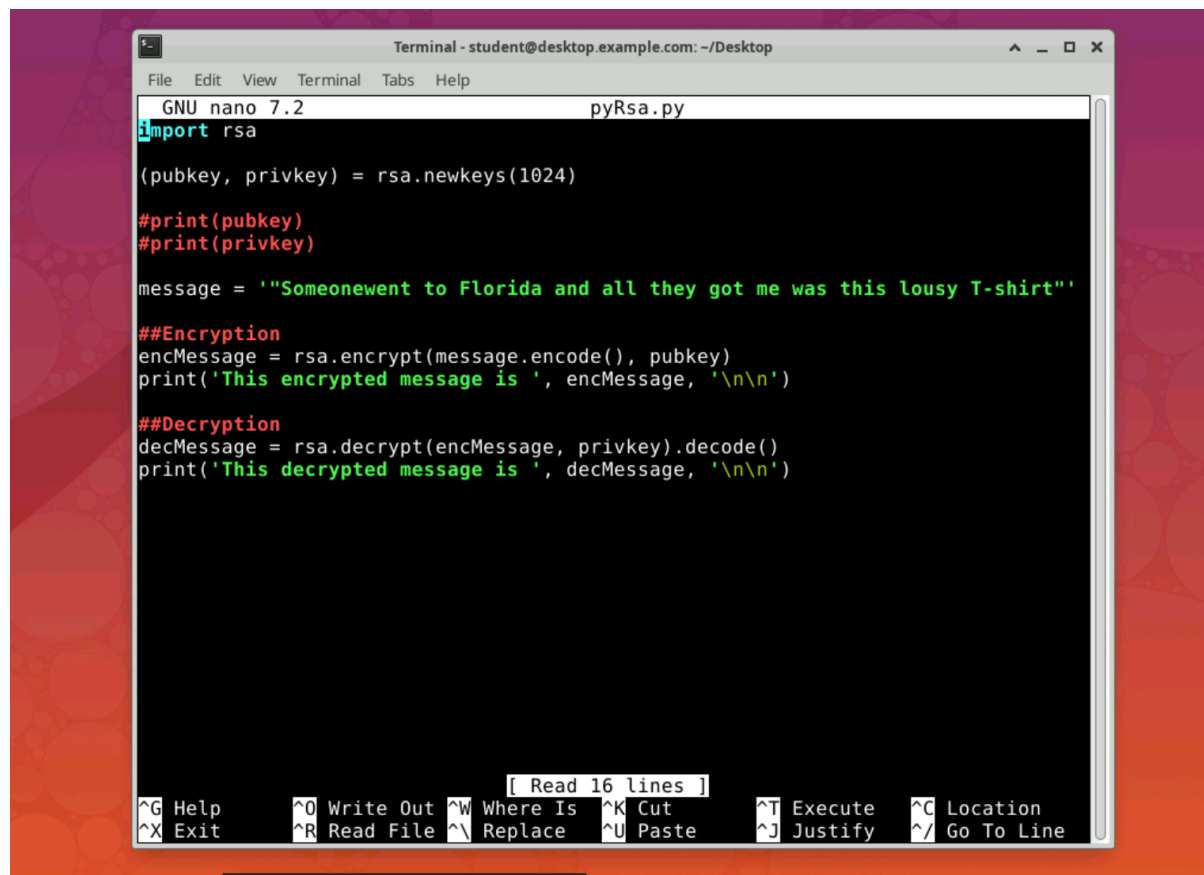
#print(pubkey)
#print(privkey)

message = "Someone went to Florida and all they got me was this lousy T-shirt"

##Encryption
encMessage = rsa.encrypt(message.encode(), pubkey)
print('The encrypted message is ', encMessage, '\n\n')

##Decryption
decMessage = rsa.decrypt(encMessage, privkey).decode()
print('The decrypted message is ', decMessage)
```

- Start a terminal window, navigate to the location of the program, and run the program as shown below.
- The program encrypts and then decrypts the message provided.



```
Terminal - student@desktop.example.com: ~/Desktop
File Edit View Terminal Tabs Help
GNU nano 7.2 pyRsa.py
import rsa

(pubkey, privkey) = rsa.newkeys(1024)

#print(pubkey)
#print(privkey)

message = "Someone went to Florida and all they got me was this lousy T-shirt"

##Encryption
encMessage = rsa.encrypt(message.encode(), pubkey)
print('This encrypted message is ', encMessage, '\n\n')

##Decryption
decMessage = rsa.decrypt(encMessage, privkey).decode()
print('This decrypted message is ', decMessage, '\n\n')

[ Read 16 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

NOTE:

- If you get an error message stating rsa is not recognized, use the command line to install rsa (i.e., `pip install rsa`)

- If the text you need to encrypt (message) is too long, the key may not work, you may have to increase the size of the key from 1024 to something longer.

```

root@cheche-virtual-machine: /home/cheche/Desktop
root@cheche-virtual-machine:/home/cheche/Desktop# python3 pyRsa

The encrypted message is b'\x7fa\xd2\xbfT\x8ahT\x0e?\xf1\x9c\xd6\xba2\xa3~\x16\x
a9\x04\xdd\xbd\x1c\xc1d'\xcc$\xdb\xec\xfa\x87\xc4*g!\x89^q\xc7\x87B\x98\x18\x00
R\x9c\xb6/\xedN\xfa7>\xd6\xb56\xb7s\xbd\xee[\xfd?\xab\xaa0m\xfa9K-h\xfa9\x7f\xfa1z\x
9b\x01\xd2\xada\xfa0\xfa1\xfa0\xa9\x01\xcb1\xbd\x17\xceZqs\x93z\xad@50\xb1\xce\x9b?
\xcc\xc0\x85\x8a\x85\x17\x0c\x9e\x03j\x97\x7x\x89S\xdd\xdc<\xc5\x99\xc5\x99\xd2
\x7f\xa2"

The decrypted message is "Someone went to Florida and all they got me was this
lousy T-shirt"

```

```

student@desktop.example.com:~/Desktop$ python3 pyRsa.py
This encrypted message is b'\xe97\x83\x80\xc2\xccP\xe6\xaa2\x88#\xd3G\xaa\xaa
W\xbb\xdb+\xe1\xab#\xe2\xcbUs\x88}=\xd6\xc6\x89\xd7\xfe\xbc\xfa\x07\xfa0\x1c\xed
\x8b\xc2\xe5m*\x91\x91\x84\xe9\\\x84Tno\xd9\x1cY\x18(\xd3N\x96j\x1bY\xc3(\xaf\x8
b\xd1\x86x\xec\xa7\x80=\xa7\xb3\x1f\xff\n\xc3\xe6\xb7\x15\xa2\xaeK+\x00\x0c\x87C
T\x00\x1aWy\xe2$9\x81\x0f\xc6\x90\x11\x01\xd3 \xf7@\x8c\x9b\xb8\x87\x06\xc5E\xfa8
\x1d'\x14\xcdS'

This decrypted message is "Someoneewent to Florida and all they got me was this
lousy T-shirt"

```

Class Exercise 4

- Comment out the message, the encryption, and decryption functions.
- Uncomment the print functions that print out the public and private keys to the screen. Print out both keys. Include the screenshot of the keys, make sure both keys can be easily identified.

Task 4B – Asymmetric key encryption using RSA and Python on Windows

Class Exercise 5

- Use the python program in task 4A to encrypt a message in Windows.
- Increase the key size from to 4096
- Print out the encrypted message, decrypted message, and the public and private keys.
- Reverse the keys (i.e. use the private key to encrypt and use the public key to decrypt). Did this produce the same results? Why or why not?
- Use the same key to encrypt and decrypt. Did this produce the same results? Why or why not?

```

pyRsa.py - C:\Users\student\Desktop\pyRsa.py (3.13.3)
File Edit Format Run Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (
Enter "help" below or click "Help" above for more information.
>>> import rsa
...
... (pubkey, privkey) = rsa.newkeys(4096)
...
... #print(pubkey)
... #print(privkey)
...
... message = "Hello"
...
... ##Encryption
... encMessage = rsa.encrypt(message.encode(), pubkey)
... print('This encrypted message is ', encMessage, '\n\n')
...
... ##Decryption
... decMessage = rsa.decrypt(encMessage, privkey).decode()
... print('This decrypted message is ', decMessage, '\n\n')

```

NOTE:

- If you get an error message stating rsa is not recognized, use the command line to install rsa (i.e., [pip install --trusted-host pypi.org --trusted-host pypi.python.org --trusted-host files.pythonhosted.org rsa](https://pypi.org/project/rsa/))

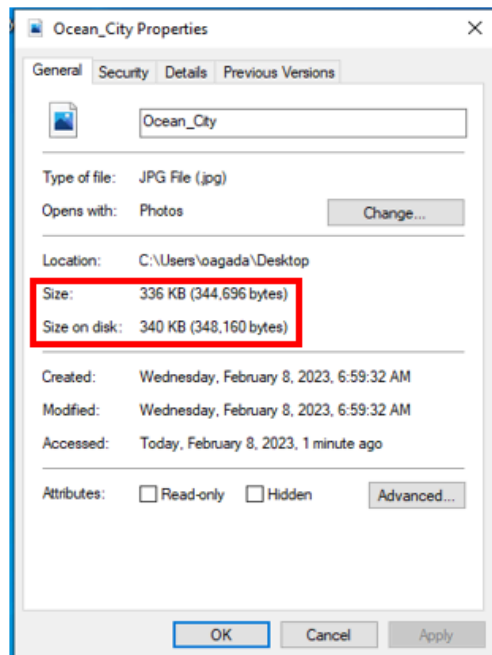
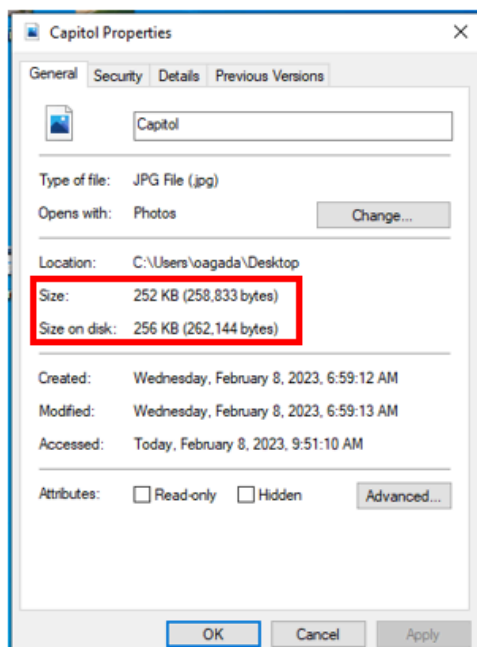
Task 5A – Steganography using the Windows command line

- Create 2 jpeg files (i.e. Capitol.jpg & Ocean_City.jpg)

**Capitol.jpg****Ocean_City.jpg**

- Record the size for each file
 capitol – 8.31 KB(8515 bytes)
 ocean city – 5.34 KB(5476 bytes)

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- Fire up the command prompt and navigate to the location of the files. Enter the following command in the console `copy /b Capitol.jpg+Ocean_City.jpg Stego.jpg`

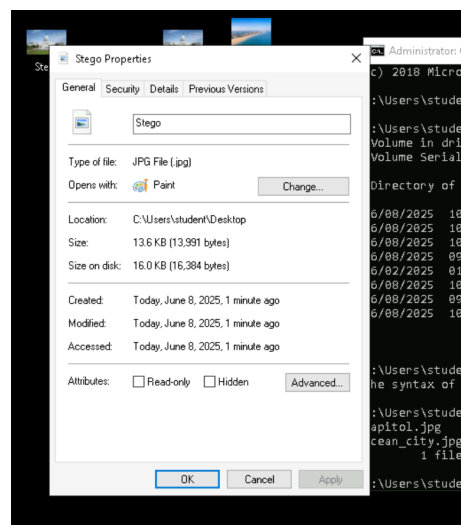
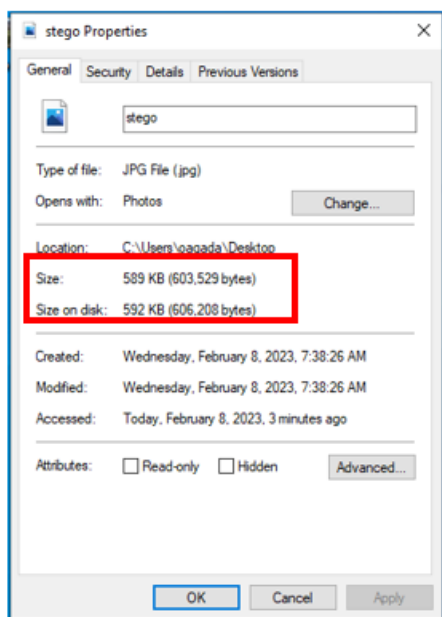
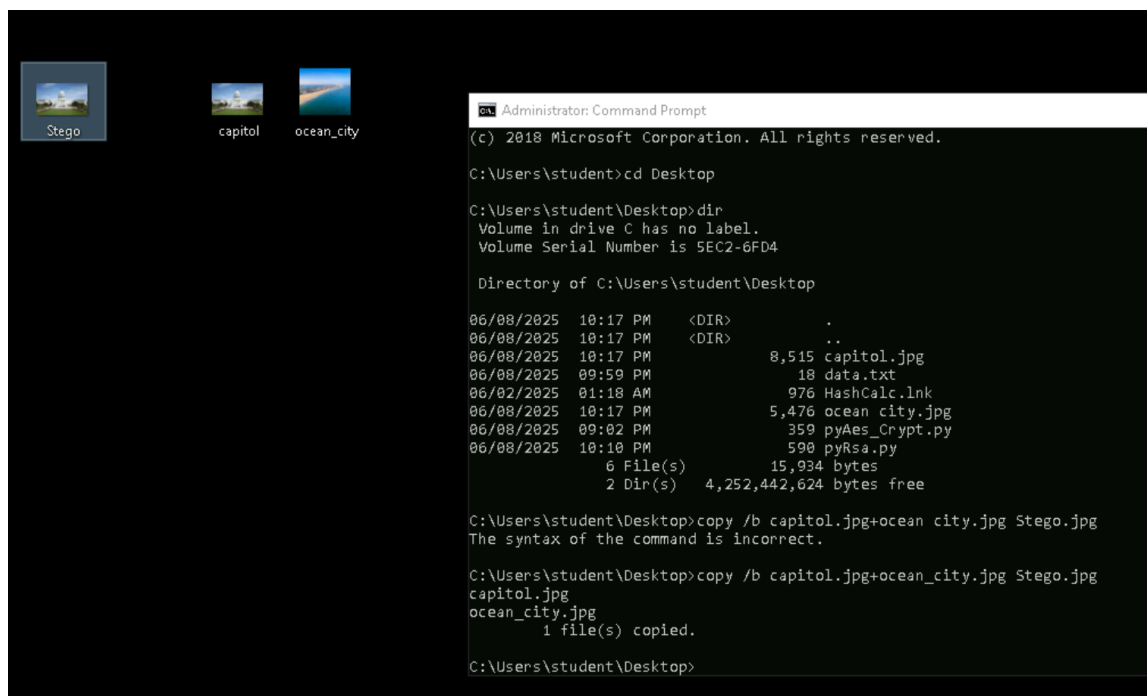
```
Command Prompt
C:\Users\oagada\Desktop>copy /b capitol.jpg+Ocean_City.jpg stego.jpg
Capitol.jpg
Ocean_City.jpg
1 file(s) copied.
```

- Capitol.jpg and Ocean_City.jpg have been combined into a third file called Stego.jpg.



Stego.jpg

- To the ordinary eyes, Stego.jpg looks exactly like Capitol.jpg but a quick check shows Stego.jpg is much larger.



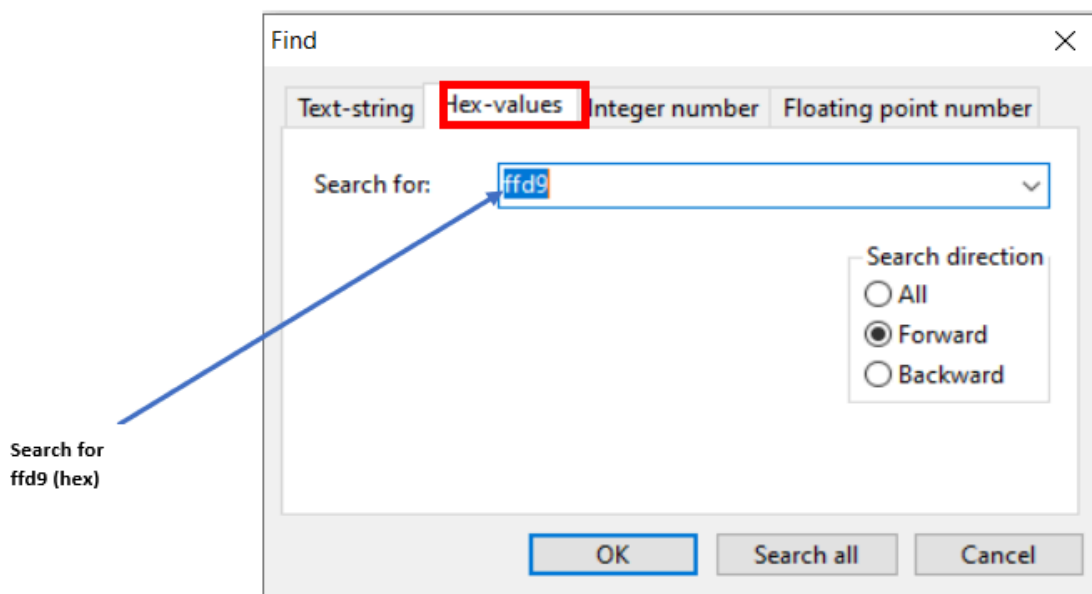
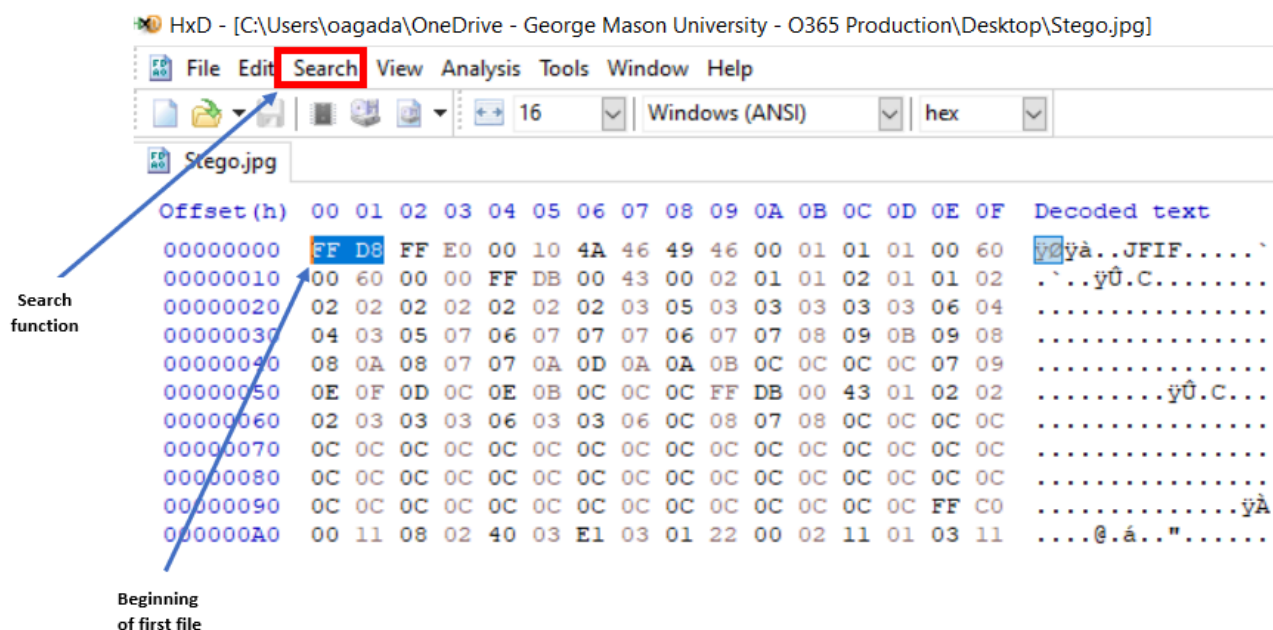
- Using forensic techniques, we can verify that stego.jpg contains 2 files

Class Exercise 6

- Download a hex viewer like HxD and extract the hidden file (i.e. Ocean_City.jpg) from Stego.jpg. Provide screenshots and itemize your steps.

TIP:

- Download HxD (<https://mh-nexus.de/en/downloads.php?product=HxD20>)
- Open the Stego file (i.e., Stego.jpg) in HxD
- All files have file signatures or magic numbers. They are unique for each file type and appear at the beginning of a file.
- The magic number for JPEG files is the hexadecimal number **ffd8**. All JPEG files begin with **ffd8** as shown in the screenshot below and end with **ffd9**.



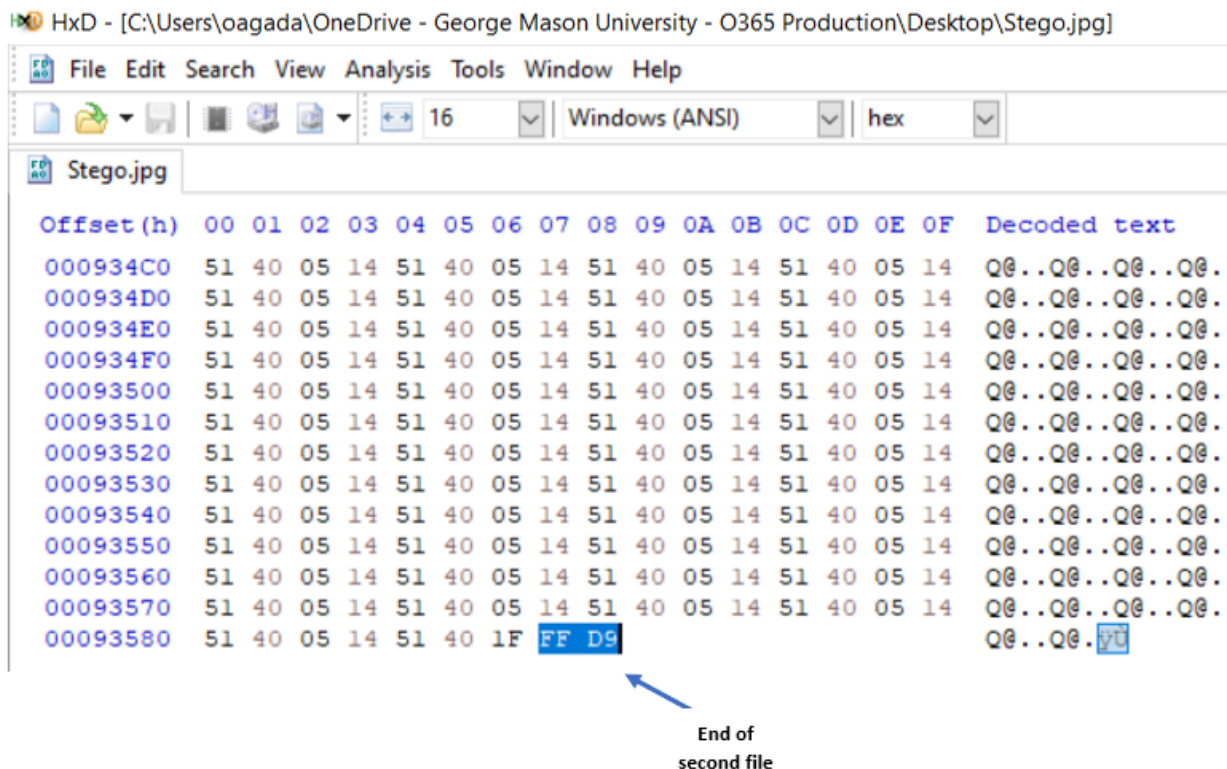
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- Use the search function in HxD to search for the end of the first file (ffd9). Remember to search for Hex-values, not Text-string, Integer, or float.

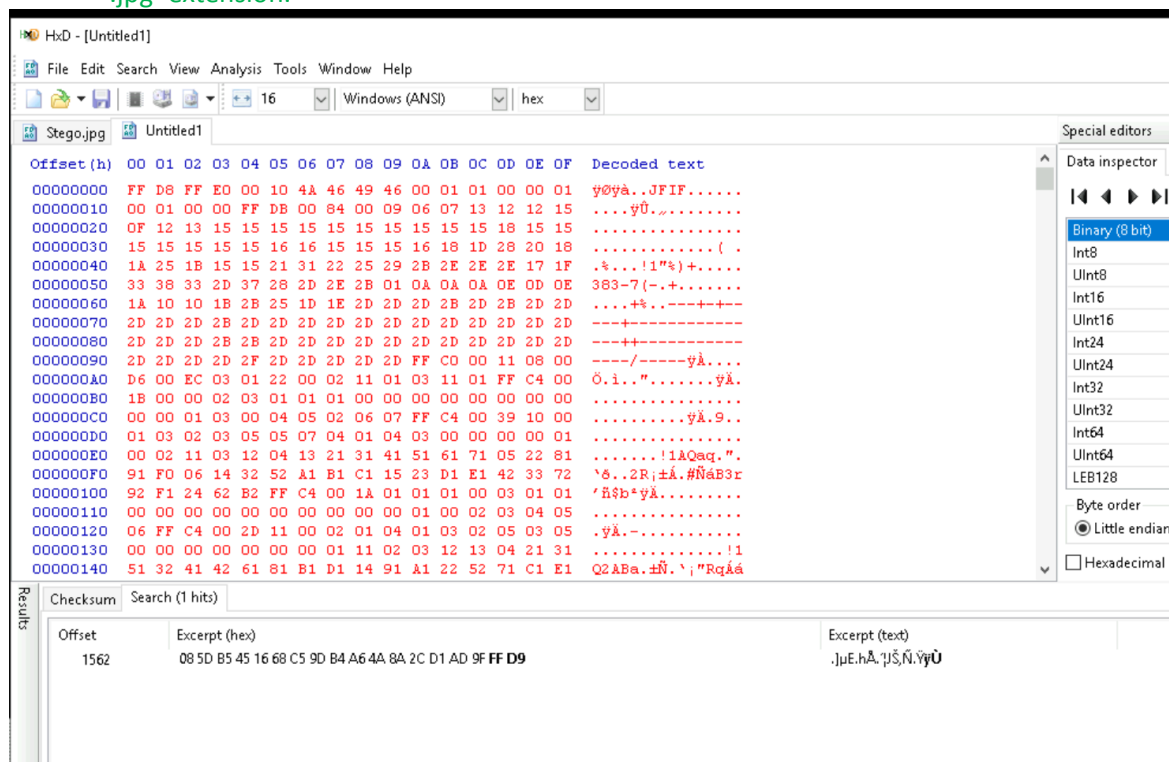
The screenshot shows the HxD hex editor interface. The main window displays the hex data of a file named 'Stego.jpg'. The hex data is organized into columns, and the decoded text is shown on the right. A red box highlights the hex value 'FF D8' at offset 0003F310, which is identified as the 'Beginning of second file'. A blue arrow points from the 'End of first file' label to the hex value '7F FE' at offset 0003F2D0.

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
0003F220	D9	F4	4F	00	5B	3C	EA	AA	F1	DA	E9	D7	11	45	C0	03	ÙòÓ. [<ë*ñÚé×.EÀ.
0003F230	85	FB	47	19	C7	38	FC	31	5F	21	D1	47	B0	A7	6E	5B	...ùG.Ç8ùl_!ÑG°Sn[
0003F240	68	4F	B5	9D	EF	73	EC	69	3F	E0	B6	DF	14	67	B5	F2	hOp.isli?àqB.guò
0003F250	A4	F0	CF	C3	89	70	30	1D	EC	2F	0B	0F	C7	ED	55	58	♠èÏÄ:po.i/..ÇiU[
0003F260	FE	1F	4B	F1	4B	C9	F2	FF	00	B0	3C	01	B7	FE	BC	6E	p.KÀKÉòý.°<.·pian
0003F270	FF	00	F9	26	BE	42	A2	AD	52	82	D9	07	B4	97	73	EB	ý.ù&Bø.R,Ù.'-sè
0003F280	BD	3F	FE	0B	45	F1	4B	4C	9B	7C	5A	17	80	81	CE	7F	¿?p.EàKL> Z.e.f.
0003F290	E3	CA	EF	FF	00	92	6A	DD	E7	FC	16	E7	E2	AD	EB	65	âÊiÿ.'jÝçü.çâ.êe
0003F2A0	BC	3D	F0	F1	4E	30	4A	D8	DE	7C	DF	5F	F4	AA	F8	E6	¼=ðñN0JØÞ ß_ð*æe
0003F2B0	8A	7C	91	EC	2E	79	1F	64	5A	7F	C1	70	3E	2C	D9	C6	Š 'i.y.dZ.Âp>,ÙE
0003F2C0	55	74	2F	87	C5	4F	50	26	17	64	5F	E9	4D	78	AF	ED	Ut/+ÂOP0.d.éMx'i
0003F2D0	65	FB	68	78	A3	E8	C6	9B	43	7F	13	65	FE	1F	D3	FF	eûhx&ðE>C..ip.ôý
0003F2E0	00	E1	1F	EB	47	D9	97	4A	82	58	55	BC	FF	00	2B	7E	.á.ùGÜ-J,XU+ÿ.+~
0003F2F0	E1	24	8F	FF	00	3C	97	18	C7	53	9C	F6	F2	1A	29	28	â\$.ÿ.<-ÇSæðð.) (
0003F300	45	4B	A4	1C	CE	D6	19	E4	0F	7A	29	F4	55	92	7F	FF	E;µ.ïÖ.â.z)øU'.Û
0003F310	D9	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	ÿÛÿà..JFIF.....
0003F320	60	00	00	00	00	FF	E1	00	22	45	78	69	66	00	00	4D	ÿÿÿà..JFIF.....
0003F330	4D	00	2A	00	00	00	08	00	01	01	12	00	03	00	00	00	M.*.....Exif...M

- Once you identify the end of the first file (ffd9), the second file starts immediately (ffd8).



- Highlight and copy from the beginning to the end of the second file.
- Under the file menu in HxD, open a new page and paste the copied text. Save it as a file with '.jpg' extension.



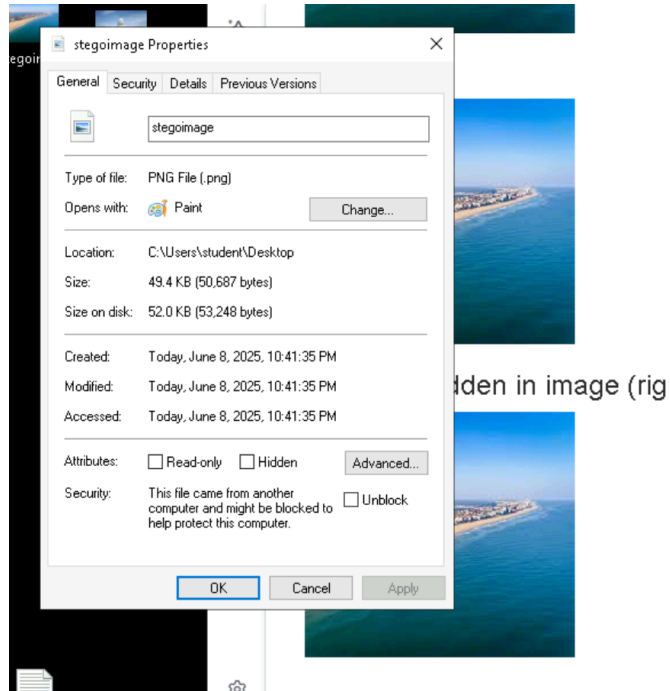
Troy Fracyon
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Task 5B – Steganography using a steganography application

- Download and install any free steganography application.

Class Exercise 7

- Use the application to hide a text file in a graphic file. Show that the operation is successful by verifying the size of the new file. Provide screenshots and itemize your steps.



Browse... ocean_city.jpg

hello

Encode

Binary representation of your message

011010000110010101101100011011000110111

Original



Normalized